Ch 6.3: Dimension Reduction - PCA

Lecture 19 - CMSE 381

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Weds, Oct 16, 2024

Announcements

Last time:

- Shrinkage: Ridge and Lasso

This lecture:

- PCA

Announcements:

- Exam #2 on Friday!
  - Bring 8.5x11 sheet of paper
  - Handwritten both sides
  - Anything you want on it, but must be your work
  - You will turn it in
  - Non-internet calculator if you want it

Lec #	Date			Reading	HW
12	Mon	9/30	Leave one out CV	5.1.1, 5.1.2	
13	Wed	10/2	k-fold CV	5.1.3	
14	Fri	10/4	More k-fold CV,	5.1.4-5	
15	Mon	10/7	k-fold CV for classification	5.1.5	
16	Wed	10/9	Subset selection	6.1	HW #4 Due Weds 10/9
17	Fri	10/11	Shrinkage: Ridge	6.2.1	
18	Mon	10/14	Shrinkage: Lasso	6.2.2	
19	Wed	10/16	~~Dimension Reduction~~ PCA	6.3	
20	Fri	10/18	~~Overflow~~, Possibly more ~~dimension reduction~~? PCR		HW #5 Due Fri 10/18
	Mon	10/21	No class - Fall break		
	Wed	10/23	**Review**		
	Fri	10/25	**Midterm #2**		

Section 1

Last time

Goal

- Fit model using all p predictors
- Aim to constrain (regularize) coefficient estimates
- Shrink the coefficient estimates towards 0

$$Y = \beta_0 + \beta_1 X_1 + \underbrace{\beta_2}_{\approx 0} X_2 + \beta_3 X_3 + \beta_4 X_4$$

Shrinkage

Find β to minimize:

Least Squares:

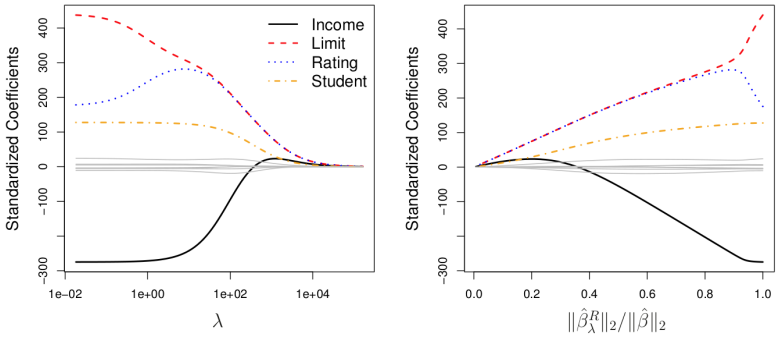
$$RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

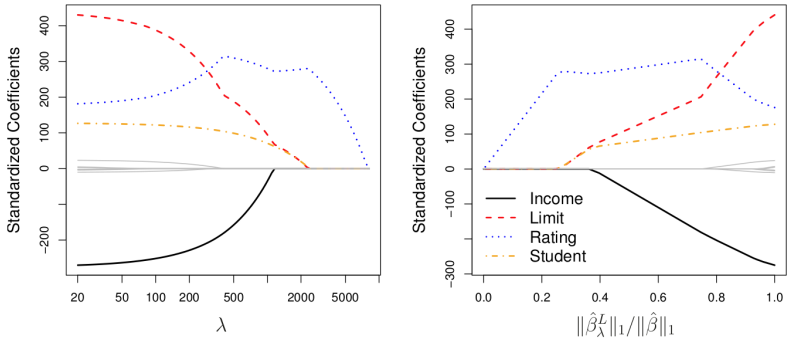
Ridge:

$$RSS + \underbrace{\sum_{j=1}^{p} \beta_j^2}$$

The Lasso:

$$RSS + \underbrace{\sum_{j=1}^{p} |\beta_j|}$$





Section 2

Dimension Reduction

Linear transformation of predictors

Original Predictors:

$$X_1, \cdots, X_p$$

New Predictors:

$$Z_1, \cdots, Z_M$$

$$Z_m = \sum_{j=1}^{p} \varphi_{jm} X_j$$

Goal: $M \ll p$

Want "good" choice of φ to do this

$$Z_m = \varphi_{1m} X_1 + \varphi_{2m} X_2 + \cdots + \varphi_{pm} X_m$$

An example or two

Two dimensions down to 1

$p = 2$ $\quad$ $M = 1$

$$Z_1 = X_1 + 3X_2$$

Matrix version

$$(Z_1) = (X_1 \ X_2)\begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

Example data

X_1	X_2	Z_1	
0	1	3	
3	4	15	$3+3(4)$
1	-1	0	$3-3$

From $p=3$ to $M=2$

$$\varphi = \begin{pmatrix} 1 & 0 \\ 0 & 1/2 \\ 0 & 1/2 \end{pmatrix}_{3\times 2}$$

$$Z_1 = X_1 + 0X_2 + 0X_3$$

$$Z_2 = 0X_1 + \tfrac{1}{2}X_2 + \tfrac{1}{2}X_3$$

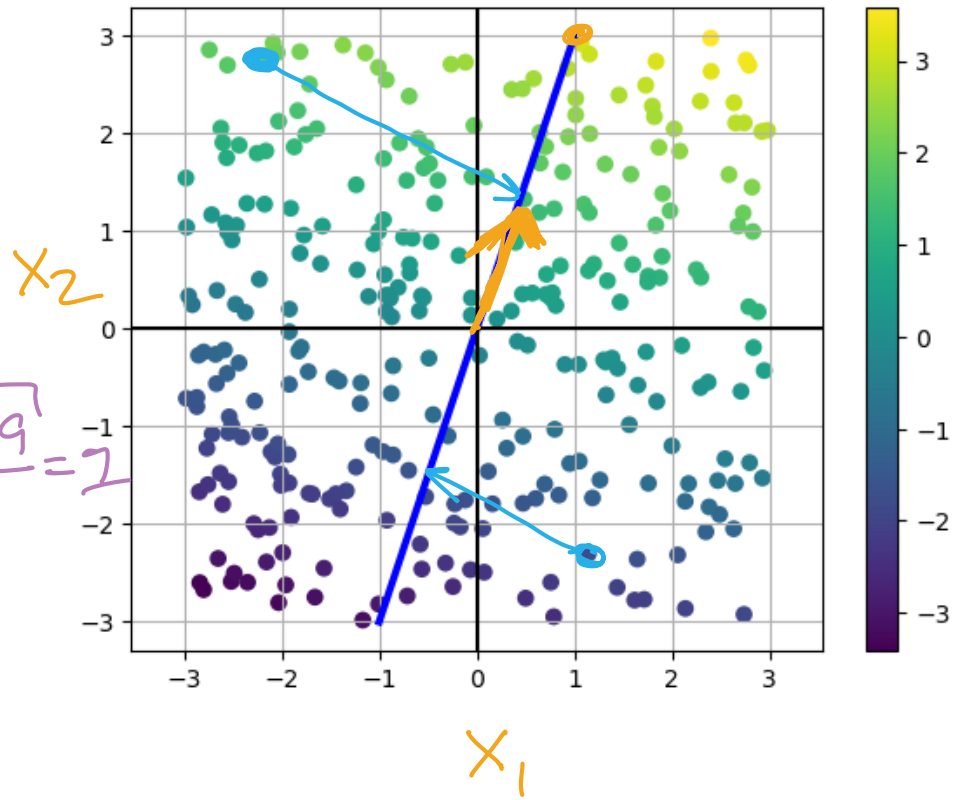
Geometric interpretation

If φ has unit norm by now

ex) $Z_1 = \frac{1}{\sqrt{10}} X_1 + \frac{3}{\sqrt{10}} X_2$

Check unit norm

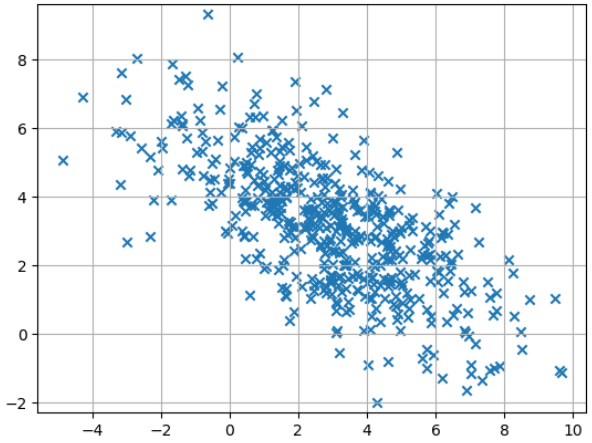
$$\sqrt{\left(\frac{1}{\sqrt{10}}\right)^2 + \left(\frac{3}{\sqrt{10}}\right)^2} = \sqrt{\frac{1+9}{10}} = 1$$

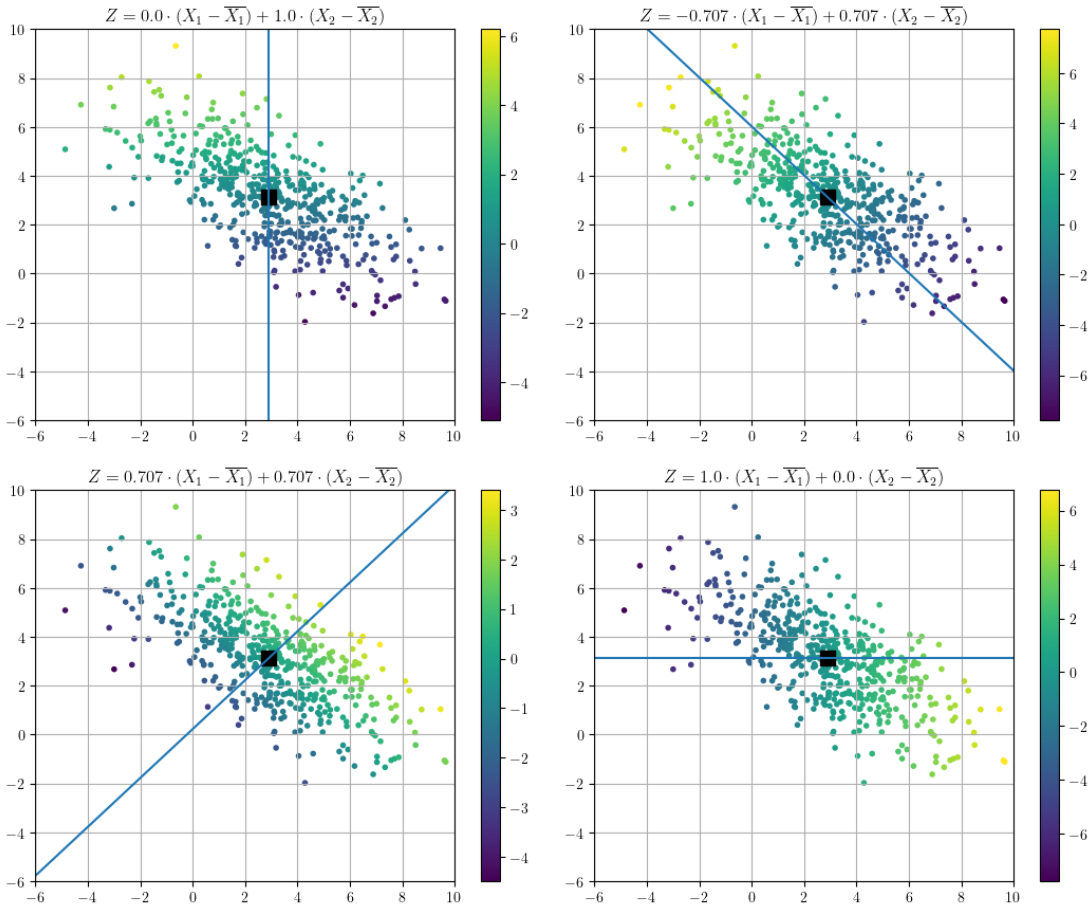


Projection onto a line

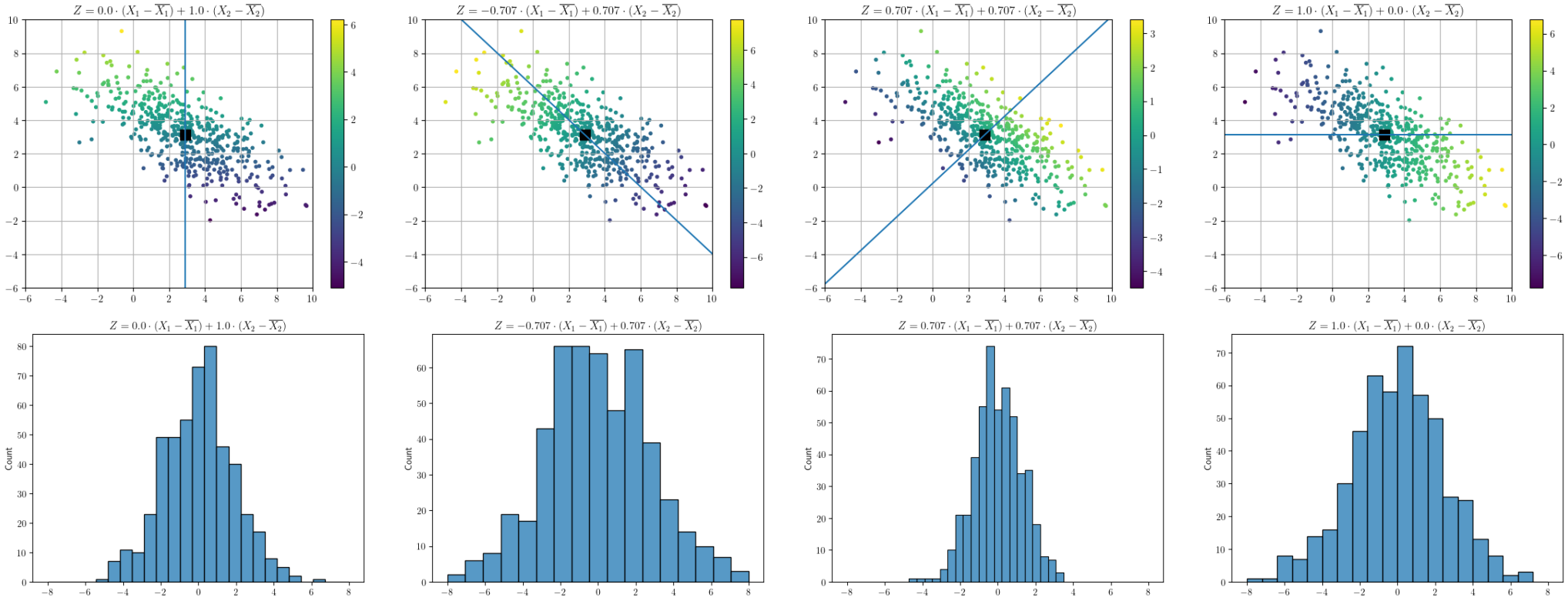
https://www.desmos.com/calculator/cih7wy8oyg

Different projections





Histograms of Z values



The goal

most variance in the projection

- Find good φ's for some $M \ll p$ ← today's job
- Fit regression model on Z_i's using least squares

$$y_i = \theta_0 + \sum_{m=1}^{M} \theta_m z_{im} + \varepsilon_i$$

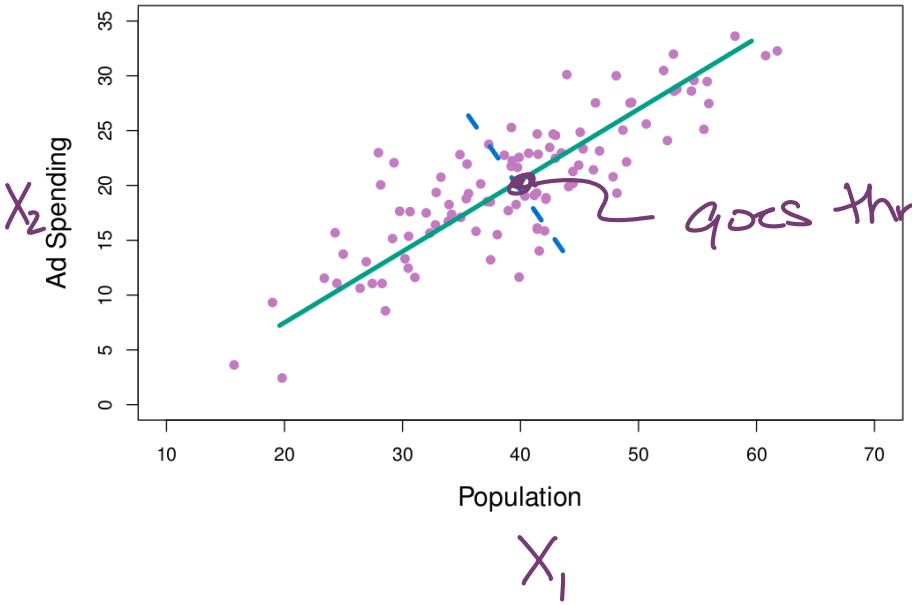
next time

Section 3

PCA

Principal component analysis

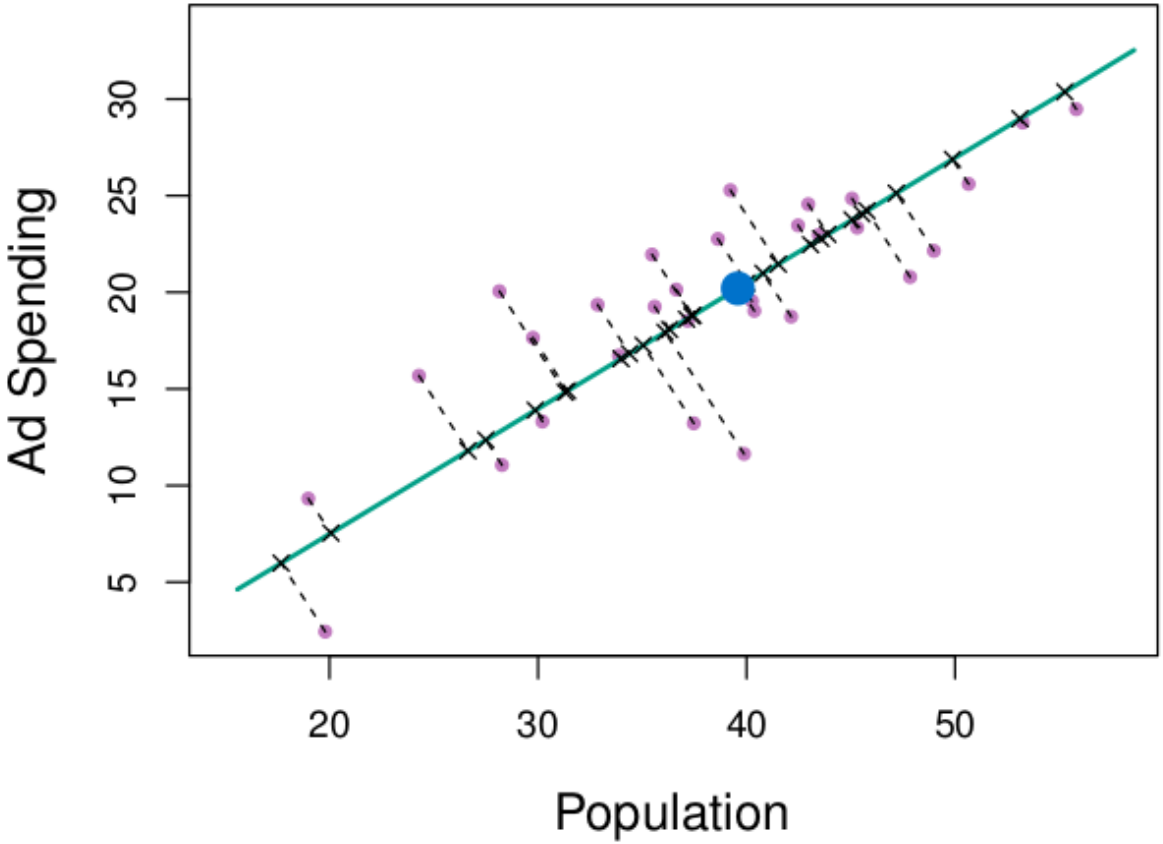
An example dataset



goes through $(\bar{X}_1, \bar{X}_2)$

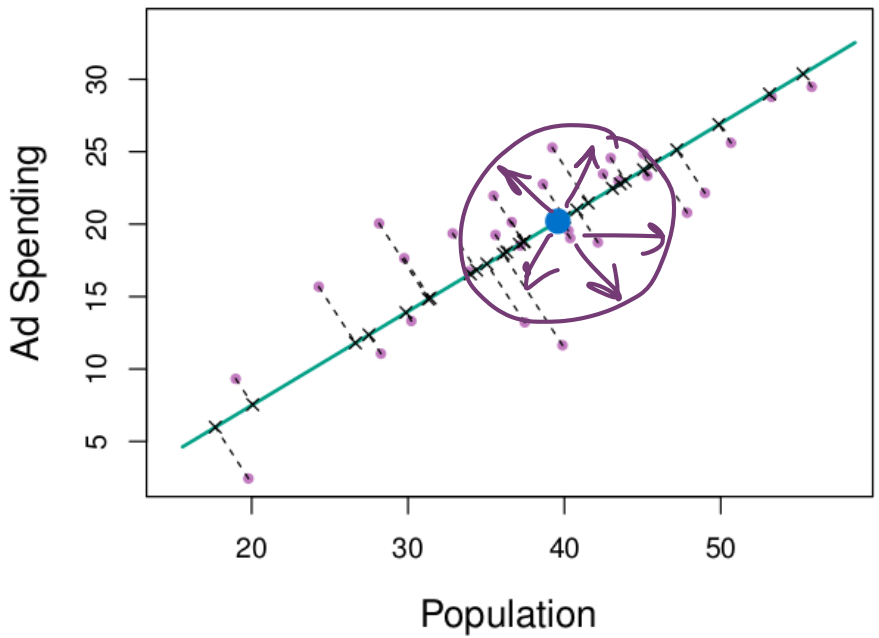
The green line is the direction of most variability

Projection onto first PC



$$Z_1 = 0.839 \cdot (\text{pop} - \overline{\text{pop}}) + 0.544 \cdot (\text{ad} - \overline{\text{ad}})$$

What does it mean to have the highest variance



Of any lin combo of pop and ad,

this choice maximizes

$$Var\left(\varphi_{11}(pop-\overline{pop}) + \varphi_{21}(ad-\overline{ad})\right)$$

Subject to

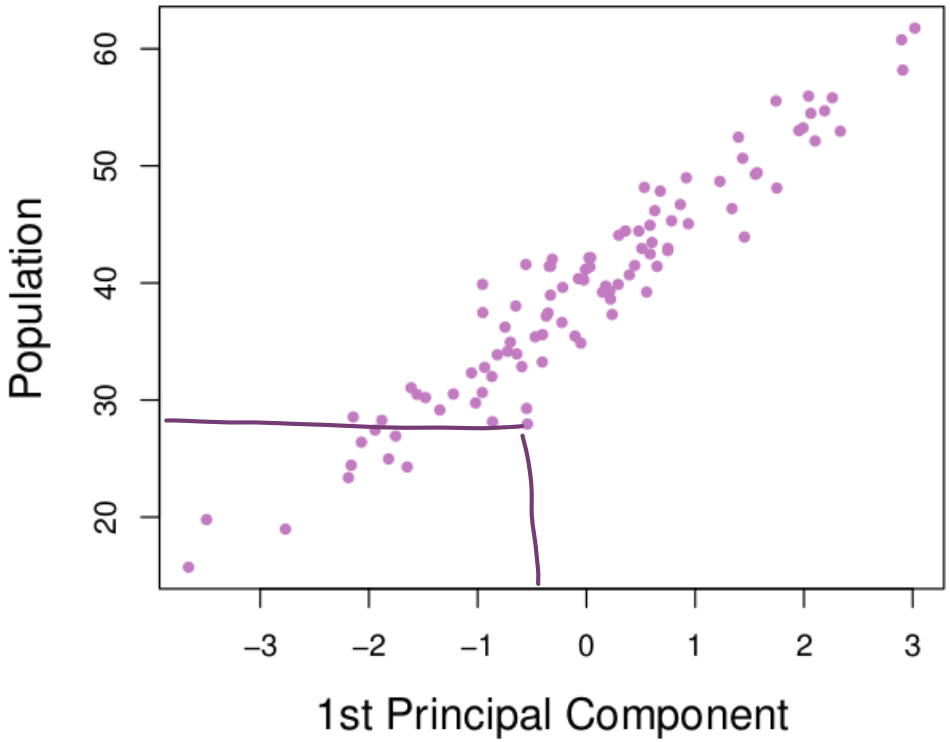
$$(\varphi_{11})^2 + (\varphi_{21})^2 = 1$$

This is called the 1st principal component (1st PC)

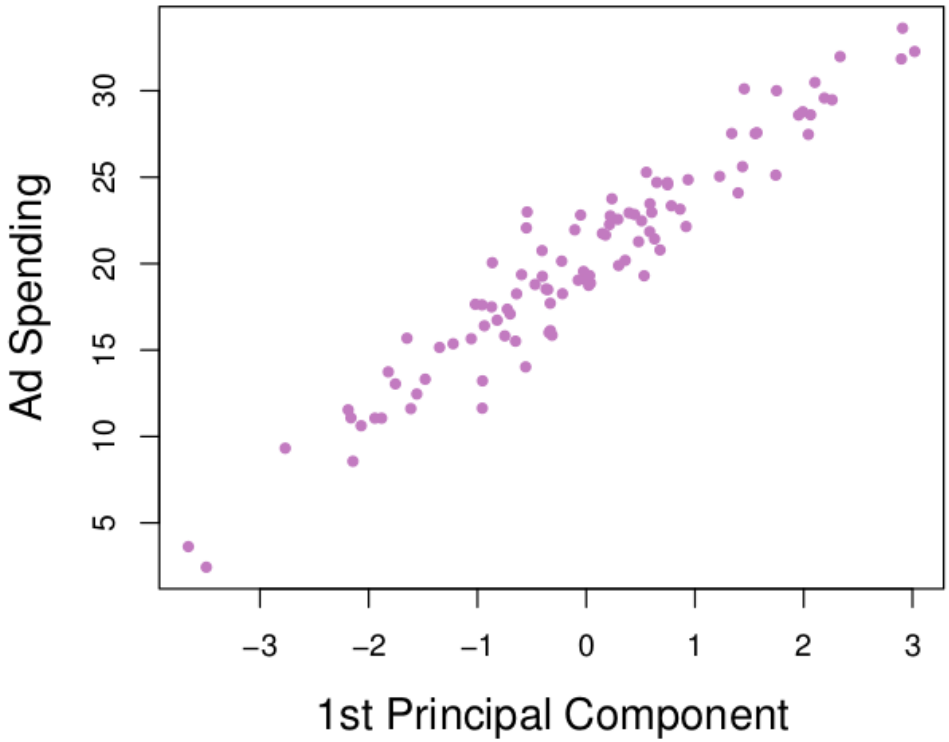
Toy for learning PCA

https://www.desmos.com/calculator/qq14tyjz0z

Principal component scores



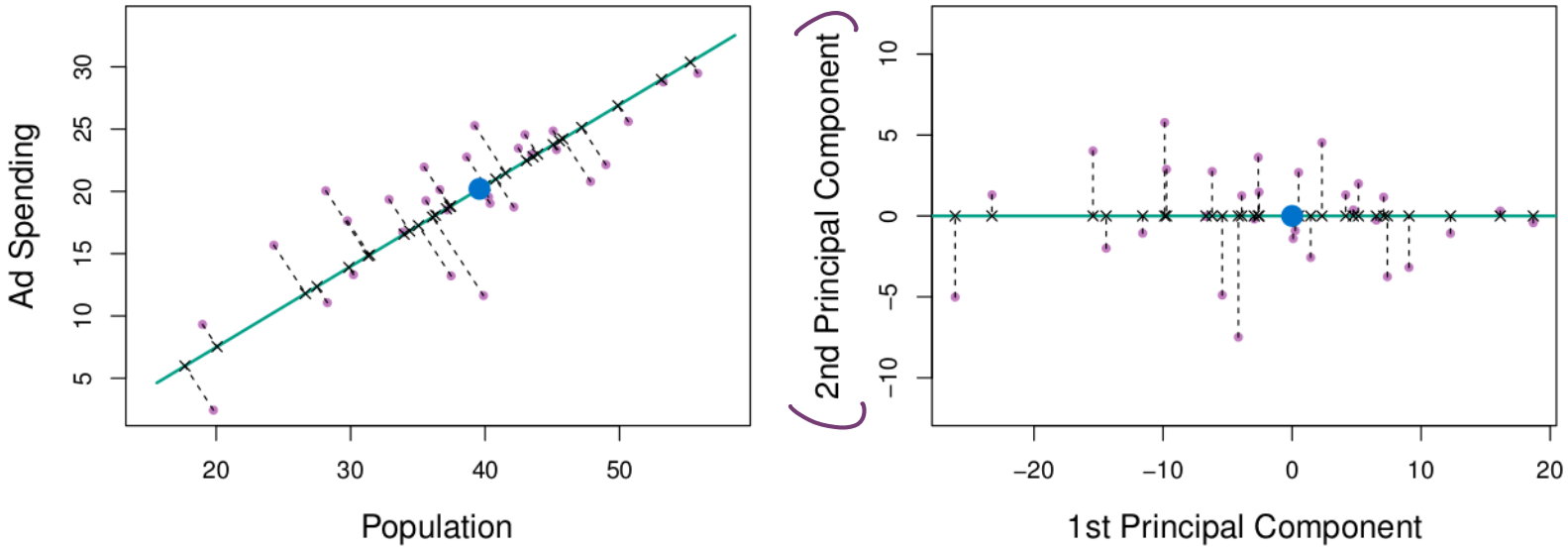
z_1



z_1

$$z_{i1} = 0.839 \cdot (\text{pop}_i - \overline{\text{pop}}) + 0.544 \cdot (\text{ad}_i - \overline{\text{ad}})$$

Another view



(Think of as rotated until next slide)

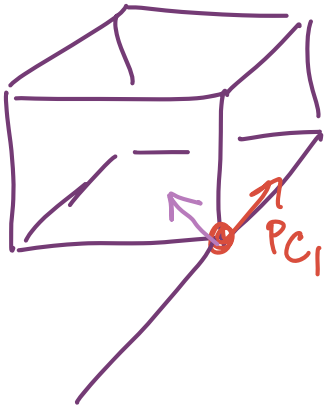
The (score) can be thought of as distance from mean
1st PC

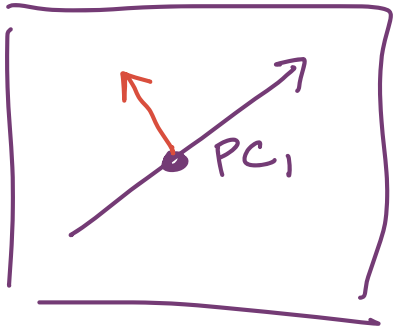
The other principal components

- 2nd PC, Z_2 linear combo of variables, uncorrelated to Z_1

  Find the one that explains the most variance

- Result is Z_2 required to be orthogonal (perpendicular)



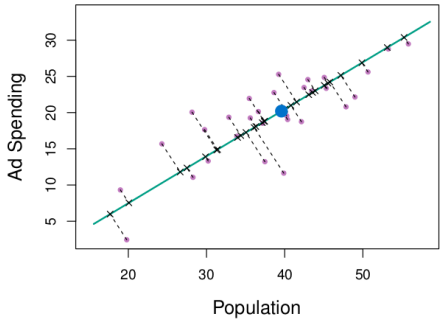


Do PCA with Penguins

TL;DR

PCA

- Unsupervised dimensionality reduction
- Choose component Z_1 in the direction of most variance using only X_i's information
- Choose Z_2 and beyond by the same method after "getting rid" of info in the directions already explained



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